

I. Fundamentals of Energy and Processes

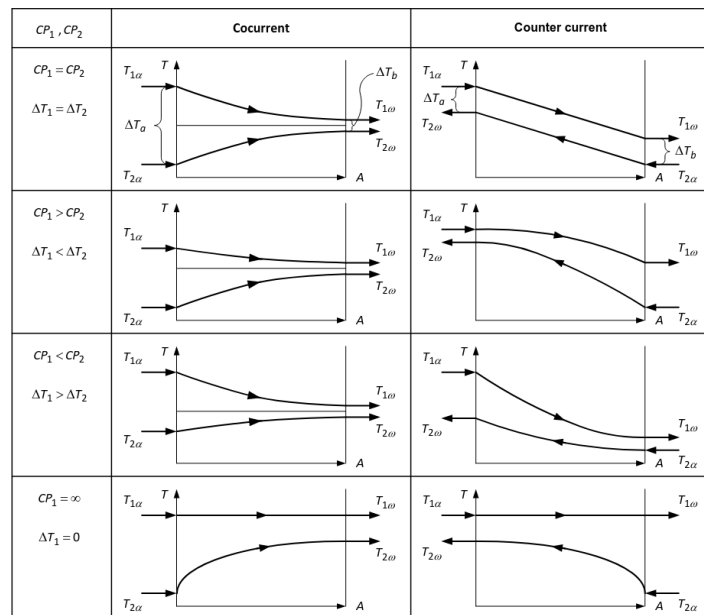
Heat exchanger

Energy balance

$$\dot{Q} = \dot{m}c_p\Delta\vartheta = CP\Delta\vartheta, \quad \dot{Q} = \dot{\phi}$$

$$\dot{m}_1c_{p1}\Delta\vartheta_1 = \dot{m}_2c_{p2}\Delta\vartheta_2 \iff CP_1\Delta\vartheta_1 = CP_2\Delta\vartheta_2$$

Average temperature difference



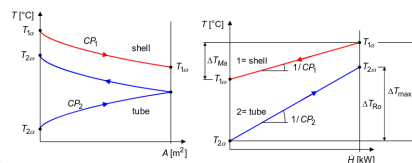
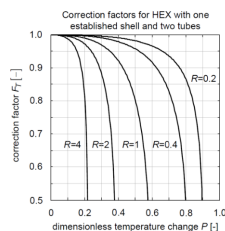
Logarithmic mean temperature difference (LMTD)

$$\Delta T_m = \frac{\Delta T_a - \Delta T_b}{\ln \frac{\Delta T_a}{\Delta T_b}}$$

$$\Delta T_a = |\vartheta_{1,a} - \vartheta_{2,a}| \quad ; \quad \Delta T_b = |\vartheta_{1,b} - \vartheta_{2,b}|$$

Corrected mean temperature heat flow

$$\dot{Q} = kA\Delta T_m \cdot F_T, \quad \text{with } 0 < F_T < 1$$



Co-current vs countercurrent flow heat exchanger

$$\dot{Q} = kA_{co}\Delta T_{m,co} = kA_{count}\Delta T_{m,count}$$

$$\frac{A_{co}}{A_{count}} = \frac{\Delta T_{m,count}}{\Delta T_{m,co}}$$

$$\frac{A_{co}}{A_{count}} = \frac{\dot{Q}_{co}}{\dot{Q}_{count}} \cdot \frac{\Delta T_{m,count}}{\Delta T_{m,co}}, \quad \text{if } \dot{Q}_1 \neq \dot{Q}_2$$

Heat capacity rate ratio (Capacity ratio)

It measures the relative ability of stream 1 vs. stream 2 to carry heat per kelvin:

$$CP_1 (\vartheta_{1,\alpha} - \vartheta_{1,\omega}) = CP_2 (\vartheta_{2,\omega} - \vartheta_{2,\alpha})$$

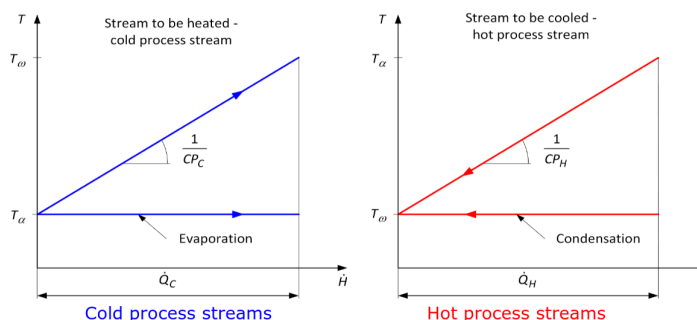
$$R_1 = \frac{CP_1}{CP_2} = \frac{\vartheta_{2,\omega} - \vartheta_{2,\alpha}}{\vartheta_{1,\alpha} - \vartheta_{1,\omega}} = \frac{\Delta\vartheta_2}{\Delta\vartheta_1}$$

T,H diagram

Cold and Hot streams

The process is extracted into material streams that must be heated (heating need) or cooled (cooling need):

Cold streams	Hot streams
Heating needed (heat sink): - Warming of milk for pasteurisation - Heating a heat transfer oil - Preparation of hot water - Evaporation of fruit juice	Cooling needed (heat source) - Cooling of a salt solution - Cooling a polymer melt - Cooling water from air compressors - Chiller refrigerant - Flue gas



Slope of the streams

$$\frac{\Delta T}{\Delta \dot{H}} \iff \frac{\Delta T_1}{CP_1 \cdot \Delta T_1} = \frac{1}{CP_1} \iff \frac{\Delta T_2}{CP_2 \cdot \Delta T_2} = \frac{1}{CP_2}$$